



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&C) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EC501(20EC603/16EC502) - INFORMATION THEORY AND ERROR CONTROL CODING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

* Marks BT* CO* PO*

- | | | | | |
|---|---|-----|---|---|
| <p>1. a) Consider a discrete memoryless source with the source alphabet $S = \{S_0, S_1, S_2\}$ with the probabilities $\{0.7, 0.15, 0.15\}$.
 (a) Calculate the entropy of the source.
 (b) Calculate the entropy of the second order extension of the source. Hence show $H(S^2) = 2H(S)$</p> | 8 | L*3 | 1 | 1 |
| <p>b) Find the minimum number of symbols 'r' in the coding alphabet for devising a prefix code such that $W = \{0, 5, 0, 5, 5\}$. Devise such a code. (W represents the set of codeword lengths) for the corresponding $L = \{1, 2, 3, 4, 5\}$.</p> | 6 | L3 | 1 | 1 |
| <p>c) List the properties of a group. Consider $G = \{0, 1, 2, 3, 4, 5, 6\}$. Test whether G is a group under addition modulo 7 and multiplication modulo 7.</p> | 6 | L3 | 2 | 1 |
| | | | | |
| <p>2. a) Find the information content of a message that consists of a digital word 9 digits long in which each digit may take on one of the five possible levels. The probability of sending any of the five levels is assumed to be equal and the level in any digit does not depend on the values taken by previous digits.</p> | 8 | L3 | 1 | 1 |
| <p>b) Encode the following using LZ algorithm.
 $W_PNW_NW_WPN_P$</p> | 6 | L3 | 1 | 1 |
| <p>c) Determine whether $f(x) = x^4 + x^3 + x^2 + x + 1$ is primitive over $F_2[x]$.</p> | 6 | L3 | 2 | 1 |
| | | | | |
| <p>3. a) Consider a discrete memoryless source with $S = \{X, Y, Z\}$ with probabilities $p = \{0.7, 0.15, 0.15\}$
 (a) Using Huffman coding algorithm, find the codewords in binary. Find the source efficiency and redundancy.
 (b) Consider the second order extension of the source. Compute the codewords for this extended source and also find its efficiency.</p> | 8 | L3 | 1 | 1 |
| <p>b) A source emits 6 symbols with the following Probability distribution $\{0.5, 0.25, 0.125, 0.0625, 0.03125, 0.03125\}$ what should be the value of average code length, in bits/symbol, that would achieve an efficiency of 100%.</p> | 6 | L3 | 1 | 1 |
| <p>c) Consider an integer group $G = \{0, 1, 2, \dots, 31\}$ under mod 32 addition. Show that $H = \{0, 4, 8, 12, 16, 20, 24, 28\}$ forms a subgroup of G. Decompose into cosets with respect to H.</p> | 6 | L3 | 2 | 1 |

Unit – II

4. a) Consider a (5,1) repetition code whose generator matrix is given by $G = [1; 11111]$
 i) find all [possible code rectors
 ii) Determine the error detecting and correcting capabilities of the code
 iii) consult its standard array
 iv) using the standard array, decode the sequence (1,1,0,1,1). 10 L3 3 1
- b) Consider a (15,7) binary cyclic code with $g(x) = 1+x^4+x^6+x^7+x^8$.
 (i) Find the code word in systematic form for the message $u(x) = x^2+x^3+x^4$. Use encoder diagram.
 (ii) Assume that the first and last bit of the code vector $V(x) = x^2+x^3+x^4$ suffer transmission errors. Find the syndrome of $V(x)$, using the syndrome calculating circuit. 10 L3 4 1
5. a) In a LBC, the syndrome is given by
 $S_1 = r_1+r_2+r_3+r_5$
 $S_2 = r_1+r_2+r_4+r_6$
 $S_3 = r_1+r_3+r_4+r_7$
 (i) Find parity check matrix
 (ii) Draw the encoder circuit.
 (iii) Find all code vectors.
 (iv) How many errors it can detect and correct?
 (v) What is the syndrome for the received data 1 0 1 1 0 1 1? 10 L3 3 1
- b) Consider a (15,5) binary cyclic code with $g(x) = 1+x+x^2+x^4+x^5+x^8+x^{10}$.
 (i) Draw the encoder circuit.
 (ii) Find the codeword for the data $u(x) = 1+x^2+x^4$ from encoder circuit.
 (iii) Draw the syndrome calculating circuit 10 L3 4 1
6. a) Consider a systematic (8,4) LBC whose parity check equations are:
 $V_5 = u_1+u_2+u_4$
 $V_6 = u_1+u_2+u_3$
 $V_7 = u_1+u_3+u_4$
 $V_8 = u_2+u_3+u_4$
 (i) Write the generator and parity check matrices.
 (ii) Find all possible code vectors.
 (iii) Find the minimum Hamming weight. 8 L3 3 1
- b) Consider a (15,11) cyclic code with $g(x) = 1+x+x^4$
 (i) Draw the encoder circuit for this code.
 (ii) Illustrate the encoding procedure for the message vector 11001101011 by listing the states of the register. 6 L3 4 1
- c) Consider a (15,7) binary cyclic code with $g(x) = 1+x^4+x^6+x^7+x^8$. Find the codeword in systematic form for the input sequence (0101010) using encoder circuit. 6 L3 4 1

Unit – III

7. a) Consider the convolution encoder shown in Fig. 7(a)

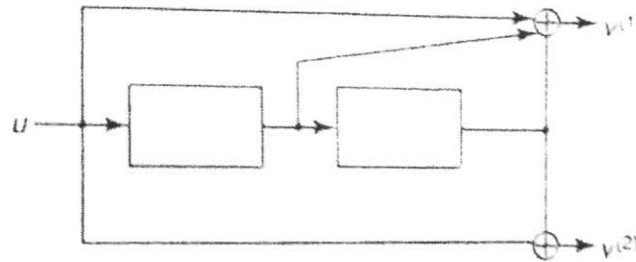


Figure.7(a)

- (i) Find the impulse responses of the encoder.
 (ii) Find the output for the message (1 0 0 1 1) using time-domain and transform-domain approach.
- b) Consider (3,1, 2) convolution encoder with $g_1 = (1 \ 1 \ 0)$, $g_2 = (1 \ 0 \ 1)$, $g_3 = (1 \ 1 \ 1)$.
 (i) Draw the encoder diagram.
 (ii) Find the code vector corresponding to the message sequence (11101) in time domain and transform domain approach.
8. a) Consider (3,1, 3) convolution code with $g_1 = (1 \ 0 \ 1 \ 1)$, $g_2 = (1 \ 1 \ 1 \ 1)$, $g_3 = (1 \ 1 \ 0 \ 1)$. Draw its encoder diagram. Also find the encoded sequence for the input (11101101) using time domain approach.
- b) Explain turbo codes.
- c) Consider (3,1,2) convolution code $g_1 = (1 \ 1 \ 0)$, $g_2 = (1 \ 0 \ 1)$, $g_3 = (1 \ 1 \ 1)$
 (i) Find the code rate
 (ii) Draw the encoder diagram
 (iii) Find the codeword for the message sequence (11101) using transform domain method.

10 L3 5 1

10 L3 5 1

8 L3 5 1

6 L2 5 1

6 L3 5 1

BT* Bloom's Taxonomy, L* Level;

CO* Course Outcome; PO* Program Outcome
